

DISHAREE BHOWMICK

Machine Learning Engineer | Data Scientist | Generative AI & Computer Vision

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PROFESSIONAL SUMMARY

Machine learning professional with 6+ years of experience: nearly 4 years at Cognizant Technology Solutions delivering data analytics, predictive modeling and generative AI for enterprise financial-services clients, followed by 3 years at Auburn University (USA) as a machine learning engineer and technical instructor. Built RAG pipelines and fine-tuned LLMs for enterprise workflows; cut recurring IT incidents by 25%; first-author publication on time-series compression (up to 30x smaller input, 85–90% accuracy retained). M.S. Computer Science (USA), GPA 3.83.

CORE SKILLS

Generative AI & LLMs: Large language models, LLM fine-tuning, Retrieval-Augmented Generation (RAG), Transformers, attention, model evaluation and benchmarking

Machine Learning: Supervised and unsupervised learning, gradient boosting and ensembles, feature engineering, data cleaning, missing-data handling, model validation and evaluation

Deep Learning & Computer Vision: PyTorch, TensorFlow, CNNs, RNNs/LSTMs/GRUs, attention, Vision-Language Models (VLMs), video analytics, event segmentation

Statistics & Time Series: Statistical tests, probability theory, time-series forecasting, change-point detection (BIC), autocorrelation, seasonality, stationarity

Data, Cloud & BI: Python (pandas, NumPy, scikit-learn), SQL, relational databases, Azure, AWS, GCP, Power BI, Tableau

PROFESSIONAL EXPERIENCE

Cognizant Technology Solutions

Pune, India

Associate | Software Engineer → Data Analyst → Machine Learning Engineer

Sep 2019 – Jun 2023

Delivered data analytics, machine learning and generative AI solutions for enterprise financial-services clients.

Generative AI (Mar 2022 – Jun 2023)

- Built and deployed generative AI solutions that integrated large language models into financial-services client workflows, working across Azure, AWS and GCP cloud environments.
- Designed and implemented Retrieval-Augmented Generation (RAG) pipelines that grounded LLM responses in enterprise knowledge bases, improving answer accuracy and reducing hallucinations.
- Fine-tuned pre-trained LLMs on domain-specific datasets and benchmarked them against base models to adapt them for enterprise use cases.
- Took solutions from prototype to deployment, collaborating with client and delivery teams to fit LLM features into existing business processes.

Machine Learning (Mar 2021 – Feb 2022)

- Led an internal predictive analytics project, owning model selection and benchmarking of supervised learning models, including gradient boosting, using Python, SQL and scikit-learn.
- Standardized preprocessing, missing-data handling, feature engineering and model validation into a repeatable workflow so every candidate model was evaluated on the same held-out data.

Data Analytics (Mar 2020 – Feb 2021)

- Reduced recurring IT incident volume by **25%** by partnering with operations teams on root-cause analysis and turning findings into targeted fixes.
- Built Python-based analytics dashboards in Power BI and Tableau on incident data queried with SQL, giving enterprise clients visibility into IT incident trends.
- Completed a 6-month structured analytics training program (Sep 2019 – Feb 2020) covering exploratory data analysis, statistical methods and data preprocessing.

Auburn University — Department of Computer Science & Software Engineering

Auburn, AL, USA

Machine Learning Engineer (Research Assistant)

May 2024 – Present

Applied ML engineering role delivering computer vision, time-series and multimodal AI solutions; reporting to Dr. Sathyanarayanan N. Aakur. Began as a graduate research assistant during the M.S.; continued full-time in the role after graduating in May 2025.

Time-Series Data Compression (STaTS)

- Designed and delivered STaTS, an unsupervised framework that compresses time-series data using BIC-based multi-scale change detection, reducing model input size by **up to 30x** while retaining **85–90%** of full-resolution accuracy.
- Validated the framework at scale across 150+ benchmark datasets (UCR, UEA, ETT), building reproducible evaluation pipelines in Python and PyTorch; results published in Pattern Recognition Letters (2026).

Vision-Language Models & Video Analytics

- Developing video analytics pipelines that adapt Vision-Language Models (VLMs) for open-world event segmentation, detecting and localizing activities outside the categories seen in training.
- Contributed to CVT-Bench, an evaluation benchmark that stress-tests the spatial reasoning of multimodal LLMs under viewpoint changes, supporting model robustness assessment.
- Contributed to a computer-vision system that estimates livestock respiration from video for smart-agriculture monitoring (IEEE MWSCAS 2026).

Auburn University — Samuel Ginn College of Engineering

Graduate Teaching Assistant — Technical Training & Instruction

Auburn, AL, USA

Aug 2023 – May 2025

- Served as **Lead Teaching Assistant** for Introduction to Algorithms (COMP 3270), coordinating the TA team, assessment workflows and student support for the course.
- Delivered technical instruction and problem-solving support for Formal Languages (COMP 4200), translating complex theory into clear, structured explanations.
- Led hands-on lab sessions as Lab Instructor for Introduction to Engineering (ENGR 1110), guiding students through practical engineering exercises.

ADDITIONAL EXPERIENCE

- **Software Development Intern**, Ogma Tech Lab, Kolkata (Jul – Aug 2018) — built the "Puja Parikrama" Android app using Android Studio and Firebase.

EDUCATION

M.S. Computer Science — Auburn University, Alabama, USA

Aug 2023 – May 2025

GPA: 3.83 | Coursework: Deep Learning, Machine Learning, NLP, Computational Imaging, Image & Video Processing

B.Tech. Computer Science — University of Engineering and Management, Kolkata, India

Aug 2015 – Jul 2019

GPA: 3.6

SELECTED PUBLICATIONS

- **Bhowmick, D.**, Ramanathan, R., Aakur, S.N. STaTS: Structure-aware temporal sequence summarization via statistical window merging. **Pattern Recognition Letters**, Vol. 202, 2026. (First author)
- Palem, R.R., Garcia-Ascolani, M.E., **Bhowmick, D.**, et al. Vision-Based Livestock Respiratory Monitoring for Smart Agriculture Sensing Systems. **IEEE MWSCAS**, 2026.
- Vellamcheti, S., Kothapalli, U.K., **Bhowmick, D.**, Aakur, S.N. CVT-Bench: Counterfactual Viewpoint Transformations Reveal Unstable Spatial Representations in Multimodal LLMs. arXiv:2603.21114, 2026.
- Palem, R.R., **Bhowmick, D.**, Garcia-Ascolani, M.E., et al. Physiological Signal-Aware Livestock Respiration Estimation from Video. Preprint, 2026.

AWARDS & RECOGNITION

- **100+ Women Strong Outstanding Departmental Annual Graduate Award** (2026), Samuel Ginn College of Engineering — one of two graduate honorees in Computer Science & Software Engineering. **Award value: \$2,000.**
- **Auburn University's sole CSSE nominee** for the ACM SIGHPC Fellowship
- **Rising Star Award** for project delivery and multiple **Star of the Month** awards — Cognizant Technology Solutions